

FROM CLASSICAL LDPC CODES TO GOOD QUANTUM LDPC CODES AND DECODING

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Physical qubits are noisy, so quantum error correction encodes k logical qubits into a larger system of n physical qubits and measures checks that reveal errors without revealing the encoded state; a *code* is the set of valid encoded strings, or in the quantum case a subspace of valid states, whose redundancy and distance make likely errors distinguishable and correctable. We begin with classical coding theory because Pauli errors have a binary description and CSS quantum codes are constructed from binary parity-check matrices.

Classical: LDPC and Tanner codes. A binary linear code is specified by a parity-check matrix $H \in \mathbb{F}_2^{m \times n}$:

$$C = \ker H \subseteq \mathbb{F}_2^n, \quad k = n - \text{rank } H, \quad d = \min_{0 \neq c \in C} \text{wt}(c). \quad (1)$$

It is LDPC when the row and column weights of H are $O(1)$, and a family is asymptotically good when $k = \Theta(n)$ and $d = \Theta(n)$ [1]. Classical Tanner code places one bit on every edge of a constant-degree bipartite graph $\Gamma = (L \sqcup R, E)$ and imposes the same fixed local code $C_0 \subseteq \mathbb{F}_2^D$ at each vertex:

$$C(\Gamma, C_0) = \{x \in \mathbb{F}_2^E : x|_{E(v)} \in C_0 \text{ for every } v\}. \quad (2)$$

For a D -regular graph with $|L| = |R| = m$, spectral expansion is measured by

$$\left| e(S, T) - \frac{D}{m} |S||T| \right| \leq \lambda_2 \sqrt{|S||T|}, \quad (3)$$

and the guiding mechanism is *local distance + global expansion $\Rightarrow$ linear distance* [2].

Quantum: CSS and qLDPC codes. Ignoring global phase, every n -qubit Pauli error is $X^a Z^b$ with $a, b \in \mathbb{F}_2^n$. A CSS code chooses sparse matrices H_X, H_Z whose rows define X - and Z -type stabilizers. Commutation, syndrome extraction, dimension, and distance are summarized by

$$H_X H_Z^T = 0, \quad s_Z = H_X b, \quad s_X = H_Z a, \quad (4)$$

$$k = n - \text{rank } H_X - \text{rank } H_Z, \quad d = \min\{d_X, d_Z\}, \quad (5)$$

$$d_Z = \min_{\substack{z \in \ker H_X \\ z \notin \text{row}(H_Z)}} \text{wt}(z), \quad d_X = \min_{\substack{x \in \ker H_Z \\ x \notin \text{row}(H_X)}} \text{wt}(x). \quad (6)$$

The set differences are essential: a stabilizer is undetected but harmless, while an undetected vector outside the stabilizer row space is a logical operator. The original qLDPC conjecture asked for bounded-weight, bounded-degree checks satisfying (4) together with $k, d = \Theta(n)$ [3].

Development of qLDPC codes.

Year	Representative development
2009	Tillich–Zémor hypergraph products achieved positive rate and $d = \Theta(\sqrt{n})$ [5].
2021	Panteleev–Kalachev constructed explicit $[[n, \Theta(n), \Theta(n)]]$ qLDPC codes, resolving the structural conjecture [6].
2022	Quantum Tanner codes gave a cleaner Tanner-style construction; efficient and linear-time decoders were proved for several good qLDPC families [7, 8, 9].
2025	Explicit lossless vertex expanders completed a more explicit expander route to good qLDPC codes with linear-time decoding [12].
2026	Linear-time encodable and decodable asymptotically good quantum channel codes were obtained; they need not be LDPC, so they address a related but distinct objective [13].

Products and the present structural frontier. For classical parity-check matrices A, B , the hypergraph product uses

$$H_X = [A \otimes I \mid I \otimes B^T], \quad H_Z = [I \otimes B \mid A^T \otimes I], \quad (7)$$

$$H_X H_Z^T = A \otimes B^T + A \otimes B^T = 0 \quad \text{over } \mathbb{F}_2.$$

If the classical scale is L , then $n = \Theta(L^2)$ and $d = \Theta(L) = \Theta(\sqrt{n})$ [5]. Lifted products quotient the two ordinary group coordinates to one relative coordinate,

$$[(g_1, g_2)] \in G^2/G \mapsto g_1^{-1}g_2 \in G, \quad |G|^2/|G| \xrightarrow{\sim} |G|, \quad (8)$$

local product expansion + Cayley expansion $\implies d = \Theta(n)$, hence $[[n, \Theta(n), \Theta(n)]]$.

Thus good sparse commuting CSS codes exist, and several families admit provably fast syndrome decoders [6, 8, 9].

Decoding as binary inference. A detector error model is a triple (H, p, L) . For an error configuration $x \in \mathbb{F}_2^N$,

$$s = Hx, \quad \ell = Lx, \quad \mathbb{P}(x) = \prod_{j=1}^N (1 - p_j)^{1-x_j} p_j^{x_j}. \quad (9)$$

Most-likely-error decoding is the weighted binary optimization problem

$$\hat{x}^{\text{MLE}}(s) \in \arg \max_{Hx=s} \mathbb{P}(x) = \arg \min_{Hx=s} \sum_{j=1}^N w_j x_j, \quad w_j = \log \frac{1-p_j}{p_j}, \quad \hat{\ell}(s) = L\hat{x}(s). \quad (10)$$

For graphlike detector models, MWPM solves this minimum-weight representative problem and is highly effective for surface codes [4, 11]. However, the decoding note's example shows that

$$Hx = Hx' \quad \text{but} \quad Lx \neq Lx', \quad \text{and success requires} \quad L(x + \hat{x}(s)) = 0. \quad (11)$$

Accordingly, a logical-class maximum-likelihood decoder compares total posterior mass,

$$\hat{\ell}_{\text{ML}}(s) \in \arg \max_{\lambda \in \mathbb{F}_2^k} \sum_{\substack{x: Hx=s \\ Lx=\lambda}} \mathbb{P}(x), \quad (12)$$

rather than only one minimum-weight error. Exact optimal stabilizer decoding is hard in general [10]; the 2026 worm decoder approximates logical-class probabilities for matchable codes by Markov-chain sampling [14].

What is solved, and what remains. The relevant advantage is coding-theoretic rather than a quantum speedup for a classical decoding algorithm: constant rate and linear distance can coexist with sparse stabilizer checks. The original qLDPC conjecture asked whether one can construct CSS matrices satisfying

$$H_X H_Z^T = 0, \quad k = \Theta(n), \quad d = \Theta(n), \quad w_{\text{row}}, w_{\text{col}} = O(1), \quad (13)$$

where w_{row} and w_{col} denote the maximum check weight and qubit degree. This structural problem is solved: explicit good qLDPC families exist [6], and efficient syndrome decoding is known for several such families [8, 9]. The remaining frontier is to combine these properties with fast encoding, noisy-syndrome decoding, circuit-level fault tolerance, local syndrome extraction, and useful logical gates. Thus, the structural code-construction problem is solved, while stronger algorithmic and physical problems remain. Recent fast encodable and decodable quantum channel codes do not by themselves close this qLDPC frontier, since they need not be LDPC [13].

Mathematical toolkit. Coding and information theory define rate, distance, and decoding objectives. Spectral and high-dimensional expansion control global distance; finite groups, group algebras, tensor and homological algebra organize the constructions; topology identifies logical operators. Probability, statistical mechanics, Markov chains, combinatorial optimization, numerical algorithms, and complexity theory govern decoding.

REFERENCES

- [1] R. Gallager, *IRE Trans. Inf. Theory* **8** (1962).
- [2] R. Tanner, *IEEE Trans. Inf. Theory* **27** (1981).
- [3] A. Calderbank and P. Shor, *Phys. Rev. A* **54** (1996); A. Steane, *Phys. Rev. Lett.* **77** (1996).
- [4] E. Dennis, A. Kitaev, A. Landahl, and J. Preskill, *J. Math. Phys.* **43** (2002).
- [5] J.-P. Tillich and G. Zémor, arXiv:0903.0566 (2009).
- [6] P. Panteleev and G. Kalachev, arXiv:2111.03654 (2021).
- [7] A. Leverrier and G. Zémor, arXiv:2202.13641 (2022).
- [8] A. Leverrier and G. Zémor, arXiv:2208.05537 (2022).
- [9] I. Dinur, M.-H. Hsieh, T.-C. Lin, and T. Vidick, arXiv:2206.07750 (2022).
- [10] P. Iyer and D. Poulin, *IEEE Trans. Inf. Theory* **61** (2015).
- [11] O. Higgott and C. Gidney, *Quantum* **9**, 1600 (2025).
- [12] J.-T. Hsieh et al., arXiv:2504.15087 (2025).
- [13] A. Wills et al., arXiv:2603.04543 (2026).
- [14] Z. Tobias, N. Breuckmann, and B. Placke, arXiv:2603.05428 (2026).